



## Effect of some acute and prophylactic antimigraine drugs on the vasodepressor sensory CGRPergic outflow in pithed rats

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### ABSTRACT

**Aims:** This study analyzed in pithed rats the effect of several acute and prophylactic antimigraine drugs on the CGRPergic vasodepressor sensory outflow, in an attempt to investigate systemic cardiovascular effects in a model unrelated to migraine.

**Main methods:** Male Wistar pithed rats were pretreated with continuous i.v. infusions of hexamethonium (2 µg/kg.min; to block autonomic outflow) and methoxamine (15–20 µg/kg.min; to maintain diastolic blood pressure at around 130 mmHg). Under these conditions, the effect of both electrical stimulation (0.56–5.6 Hz; 50 V and 2 ms) of the spinal cord (T<sub>9</sub>–T<sub>12</sub>) or i.v. bolus injections of exogenous α-CGRP (0.1–1 µg/kg) were studied in animals pretreated with continuous i.v. infusions of sumatriptan (1–100 µg/kg.min), ergotamine (0.18–0.56 µg/kg.min), dihydroergotamine (1–10 µg/kg.min), magnesium valproate (1000–1800 µg/kg.min), propranolol (100–300 µg/kg.min) or their respective vehicles.

**Key findings:** Electrical stimulation of the spinal cord and i.v. bolus injections of exogenous α-CGRP resulted in, respectively, frequency- and dose-dependent decreases in diastolic blood pressure without affecting heart rate. Moreover, the infusions of sumatriptan, ergotamine and dihydroergotamine, but not of magnesium valproate, propranolol or their respective vehicles, dose-dependently inhibited the vasodepressor responses to electrical stimulation. In contrast, sumatriptan (10 µg/kg.min), ergotamine (0.31 µg/kg.min) and dihydroergotamine (3 µg/kg.min) failed to inhibit the vasodepressor responses to exogenous α-CGRP.

**Significance:** The above findings suggest that the acute (rather than the prophylactic) antimigraine drugs attenuate the vasodepressor sensory outflow mainly by prejunctional mechanisms. This may be of particular relevance when considering potential cardiovascular adverse effects by acute antimigraine drugs.

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### Introduction

Resistance blood vessels are innervated by sympathetic (see Westfall and Westfall, 2006a) and primary sensory (Taguchi et al., 1992a,b) nerves which modulate the vascular tone. C-fibers are primary sensory nerves originating from the spinal cord (Julius and Basbaum, 2001) and, upon stimulation, cause a non-adrenergic, non-cholinergic (NANC) vasodilatation via the release of vasodilator neurotransmitters, particularly calcitonin gene-related peptide (CGRP; Kawasaki et al., 1988). CGRP is a polypeptide of 37 amino acids predominantly located in sensory neurons and perivascular nerves surrounding blood vessels (Suzuki et al., 1989), where it is colocalized with other neuropeptides, such as substance P and neurokinin A (Van Rossum et al., 1997). Indeed, Taguchi et al. (1992b) have shown that electrical stimulation of the thoracic (T<sub>9</sub>–T<sub>12</sub>) spinal

cord in pithed rats receiving continuous infusions of hexamethonium (to block autonomic outflow) and methoxamine (for a sustained increase in systemic blood pressure) caused vasodepressor responses through the CGRP release from primary sensory nerves on the systemic vasculature. Thus, the above NANC vasodepressor responses were: (i) blocked by tetrodotoxin or capsaicin; (ii) unaffected after atropine, propranolol or pyrilamine+cimetidine; and (iii) antagonized by CGRP<sub>(8–37)</sub>, a CGRP receptor antagonist.

On the other hand, several lines of evidence suggest that antimigraine drugs such as sumatriptan (Williamson et al., 1997; Limmroth et al., 2001; Gupta et al., 2006) and dihydroergotamine (Buzzi et al., 1991) can inhibit the neurogenic vasodilatation produced by trigeminal release of CGRP. Nevertheless, to the best of our knowledge, no study has analyzed whether antimigraine drugs can inhibit the systemic vasodepressor responses produced by stimulation of the NANC perivascular sensory outflow, a mechanism that may be associated with adverse cardiovascular events of acute antimigraine drugs (Vanmolkot and de Hoon, 2006). On this basis, the present study set out to investigate in pithed rats whether acute (i.e. sumatriptan, ergotamine and dihydroergotamine) and prophylactic (i.e. magnesium

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valproate and propranolol) antimigraine drugs are capable of inhibiting the vasodepressor responses induced by stimulation of the perivascular sensory outflow (Taguchi et al., 1992b; Villalón et al., 2008) and, if so, whether they also inhibit those by i.v. bolus injections of exogenous  $\alpha$ -CGRP.

## Materials and methods

### Animals

Male Wistar normotensive rats (300–350 g) were maintained at a 12/12-h light–dark cycle (with light beginning at 7 a.m.) and housed in a special room at constant temperature ( $22 \pm 2^\circ\text{C}$ ) and humidity (50%), with food and water freely available in their home cages. All animal procedures and the protocols of the present investigation were approved by our Institutional Ethics Committee on the use of animals in scientific experiments.

### General methods

Experiments were carried out in a total of 132 rats. After anesthesia with diethyl ether and cannulation of the trachea, the rats were pithed by inserting a stainless-steel rod through the orbit and foramen magnum into the vertebral foramen (Gillespie et al., 1970). Then, the animals were artificially ventilated with room air using a model 7025 Ugo Basile pump (56 strokes/min and a stroke volume of 20 ml/kg), as previously established by Kleinman and Radford (1964). After cervical bilateral vagotomy, catheters were placed in: (i) the left femoral vein, for the continuous infusion of hexamethonium; (ii) the right femoral vein, for the continuous infusion of methoxamine; (iii) the left jugular vein, for the continuous infusions of the different antimigraine drugs; and (iv) the right jugular vein, for the bolus injections of gallamine or exogenous  $\alpha$ -CGRP. In addition, the left carotid artery was connected to a Grass pressure transducer (P23 XL), for the recording of arterial blood pressure. Heart rate was measured with a tachograph (7P4, Grass Instrument Co., Quincy, MA, U.S.A.) triggered from the blood pressure signal. Both blood pressure and heart rate were recorded simultaneously by a model 7 Grass polygraph (Grass Instrument Co., Quincy, MA, U.S.A.). At this point, the 132 rats were divided into two main sets, so that the effects produced by continuous i.v. infusions of either the vehicles (i.e. saline or propylene glycol 10% in saline) or the antimigraine drugs (i.e. sumatriptan, ergotamine, dihydroergotamine, magnesium valproate and propranolol) could be investigated on the vasodepressor responses induced by electrical stimulation of the perivascular sensory outflow (set 1;  $n=96$ ) or i.v. bolus injections of exogenous  $\alpha$ -CGRP (set 2;  $n=36$ ). The vasodepressor stimulus–response curves and dose–response curves elicited by, respectively, electrical stimulation and exogenous  $\alpha$ -CGRP were completed in about 60 min, and each response was elicited under unaltered values of resting blood pressure. The electrical stimuli (0.56, 1, 1.8, 3.1 and 5.6 Hz) as well as the dosing with  $\alpha$ -CGRP (0.1, 0.18, 0.31, 0.56 and 1  $\mu\text{g/kg}$ ) were given using a sequential schedule, in 0.25 log unit increments, at 5–10 min intervals (see below). It is noteworthy that a sequential schedule was used to facilitate the comprehension of our results, considering that the randomization of frequencies of stimulation did not modify the intensity of the vasodepressor responses (not shown). Additionally, the body temperature of each pithed rat was maintained at  $37^\circ\text{C}$  by a lamp and monitored with a rectal thermometer.

### Experimental protocols

#### Protocol 1. Electrical stimulation of the perivascular (vasodepressor) sensory outflow

In the first set of rats ( $n=96$ ), the pithing rod was replaced by an electrode enameled except for 1.5 cm length 9 cm from the tip, so that

the uncovered segment was situated at T<sub>9</sub>–T<sub>12</sub> of the spinal cord; an indifferent electrode was placed dorsally (Taguchi et al., 1992b). Prior to electrical stimulation, the animals received (i.v.): (i) a bolus injection of gallamine (25 mg/kg) to avoid electrically-induced muscular twitching; and (ii) 10 min later, a continuous infusion of hexamethonium (2  $\mu\text{g/kg.min}$ ) to block electrically-induced vasopressor responses that are produced by stimulation of the preganglionic sympathetic vasopressor outflow (Villalón et al., 1998). After 10 min, baseline values of diastolic blood pressure (a more accurate indicator of peripheral vascular resistance) and heart rate were determined and then the animals received a continuous i.v. infusion of methoxamine ( $\alpha_1$ -adrenoceptor agonist; 15–20  $\mu\text{g/kg.min}$ ). Twenty minutes later, the values of diastolic blood pressure and heart rate were determined again; under these conditions, a sustained increase in diastolic blood pressure (maintained at around 130 mmHg; see Results section) was produced. Then, this set of rats was divided into eight treatment groups (during the infusion of methoxamine) that received continuous i.v. infusions of, respectively: (i) nothing (control group,  $n=6$ ); (ii) saline (0.02 ml/min,  $n=6$ ); (iii) propylene glycol 10% in saline (0.02 ml/min,  $n=6$ ); (iv) sumatriptan (1, 10 and 100  $\mu\text{g/kg.min}$ ,  $n=18$ ); (v) ergotamine (0.18, 0.31 and 0.56  $\mu\text{g/kg.min}$ ,  $n=18$ ); (vi) dihydroergotamine (1, 3 and 10  $\mu\text{g/kg.min}$ ,  $n=18$ ); (vii) magnesium valproate (1000 and 1800  $\mu\text{g/kg.min}$ ,  $n=12$ ); and (viii) propranolol (100 and 300  $\mu\text{g/kg.min}$ ,  $n=12$ ). Ten minutes later, the values of diastolic blood pressure and heart rate were determined once again in each group; subsequently, the perivascular sensory outflow was electrically stimulated during the infusions of the above compounds (methoxamine and the subsequent treatment) to elicit vasodepressor responses by applying 10-s trains of monophasic, rectangular pulses (2 ms, 50 V), at increasing frequencies of stimulation (0.56, 1, 1.8, 3.1 and 5.6 Hz), as previously reported (Villalón et al., 2008). When diastolic blood pressure had returned to baseline levels, the next frequency was applied; this procedure was systematically performed until the stimulus–response curve had been completed. It is to be noted that only one stimulus–response curve was carried out per animal since tachyphylaxis of the vasodepressor responses was observed when eliciting a second stimulus–response curve (Villalón et al., 2008).

#### Protocol 2. Administration of exogenous $\alpha$ -CGRP

The second set of rats ( $n=36$ ) was prepared as described above, but the pithing rod was left throughout the experiment and the administration of both gallamine and hexamethonium was omitted. After a stable haemodynamic condition for at least 30 min, baseline values of diastolic blood pressure and heart rate were determined. Subsequently, the animals received a continuous i.v. infusion of methoxamine (15–20  $\mu\text{g/kg.min}$ ). Twenty minutes later, the values of diastolic blood pressure and heart rate were determined again. Then, this set of rats was divided into six treatment groups that received continuous i.v. infusions of, respectively: (i) nothing (control group,  $n=6$ ); (ii) saline (0.02 ml/min,  $n=6$ ); (iii) propylene glycol 10% in saline (0.02 ml/min,  $n=6$ ); (iv) sumatriptan (10  $\mu\text{g/kg.min}$ ,  $n=6$ ); (v) ergotamine (0.31  $\mu\text{g/kg.min}$ ,  $n=6$ ); and (vi) dihydroergotamine (3  $\mu\text{g/kg.min}$ ,  $n=6$ ). Ten minutes later, the values of diastolic blood pressure and heart rate were determined once again in each group; subsequently, the vasodepressor responses elicited by i.v. bolus injections of exogenous  $\alpha$ -CGRP (0.1, 0.18, 0.31, 0.56 and 1  $\mu\text{g/kg}$ ) were examined during the infusions of the above compounds (methoxamine and the subsequent treatment).

#### Other procedures applying to protocols 1 and/or 2

The doses of methoxamine, hexamethonium and the antimigraine drugs were continuously infused at a rate of 0.02 ml/min by a WPI model sp100i pump (World Precision Instruments Inc., Sarasota, FL, U.S.A.). Moreover, the electrical stimuli and the doses of methoxamine and  $\alpha$ -CGRP were selected on the basis of a previous study (Villalón et al., 2008) whereas the doses of the antimigraine drugs were chosen from

preliminary experiments. The interval between the different frequencies of stimulation/doses of  $\alpha$ -CGRP depended on the duration of the resulting vasodepressor responses (between 5 and 15 min), as in each case we waited until diastolic blood pressure had returned to baseline values.

## Drugs

Apart from the anesthetic (diethyl ether), the compounds used in this study (obtained from the sources indicated) were: gallamine triethiodide, hexamethonium chloride, methoxamine hydrochloride, rat  $\alpha$ -CGRP and propranolol hydrochloride (Sigma Chemical Co., St. Louis, MO, U.S.A); ergotamine tartrate and dihydroergotamine tartrate (gift: Sandoz de México, Mexico City, Mexico); magnesium valproate (gift: Armstrong Laboratorios de México, Mexico City, Mexico); sumatriptan succinate (gift: GlaxoSmithKline, Harlow, Essex, U.K.). All compounds were dissolved in saline, except ergotamine and dihydroergotamine, which were dissolved in 10% of propylene glycol in saline. These vehicles had no effect on the baseline values of diastolic blood pressure or heart rate (data not shown). Fresh solutions were prepared for each experiment. The doses of all compounds mentioned in the text refer to their respective free base.

## Data presentation and statistical evaluation

All data in the text, tables and figures, unless otherwise stated, are presented as mean  $\pm$  s.e.m. The peak changes in diastolic blood pressure by electrical stimulation or exogenous  $\alpha$ -CGRP were expressed as percent change from baseline. The difference in the absolute values of diastolic blood pressure and heart rate within one subgroup of animals before and after the corresponding treatment (see Table 1) was evaluated with paired Student's *t*-test. Moreover, a one-way analysis of variance was used to compare the absolute values of diastolic blood pressure and heart rate obtained in the different subgroups before and during the continuous infusions of methoxamine (after 20 min). Finally, the vasodepressor responses induced by electrical stimulation or exogenous  $\alpha$ -CGRP in the different subgroups of animals were compared with a two-way analysis of variance (randomized block design). The one- and two-way analyses of variance were followed, if applicable, by the Student–Newman–

Keuls' post hoc test (Steel and Torrie, 1980). Statistical significance was accepted at  $P < 0.05$ .

## Results

### Haemodynamic variables

The baseline values of diastolic blood pressure and heart rate in the 132 pithed rats were, respectively,  $55 \pm 3$  mmHg and  $252 \pm 8$  beats/min. These variables remained unchanged after gallamine or hexamethonium (see Fig. 1), but were maximally increased ( $P < 0.05$ ) during the continuous i.v. infusion of methoxamine ( $15\text{--}20$   $\mu\text{g/kg.min}$ , after 20 min) to, respectively,  $131 \pm 2$  mmHg and  $326 \pm 18$  beats/min ( $n = 132$ ). Tables 1 and 2 show the corresponding increases (absolute values) in these variables for each subgroup receiving, respectively, electrical stimulation and i.v. bolus injection of  $\alpha$ -CGRP (see the data “before”). For example, in the control group ( $n = 6$ ) the diastolic blood pressure (baseline value:  $54 \pm 2$  mmHg) and heart rate (baseline value:  $254 \pm 11$  beats/min) were significantly increased ( $P < 0.05$ ) to, respectively,  $121 \pm 8$  mmHg and  $286 \pm 15$  beats/min (see Table 1). Similar results (with significant increases) were obtained in the rest of subgroups receiving electrical stimulation or i.v. bolus injections of  $\alpha$ -CGRP, as previously reported (Villalón et al., 2008). Moreover, it is noteworthy that the absolute values of diastolic blood pressure and heart rate obtained during the continuous i.v. infusion of methoxamine before and 10 min after the different treatments had been started in the different subgroups receiving electrical stimulation or i.v. bolus injection of  $\alpha$ -CGRP were not significantly different ( $P > 0.05$ ) (see Tables 1 and 2). Except when constructing the stimulus–response curves (see below), these increases in diastolic blood pressure and heart rate were sustained throughout the experiments. Furthermore, as shown in Tables 1 and 2, diastolic blood pressure and heart rate remained without significant changes ( $P > 0.05$ ) during the infusion of the antimigraine drugs or the corresponding vehicles.

### Vasodepressor responses produced by electrical stimulation or i.v. bolus injections of exogenous $\alpha$ -CGRP

During the continuous infusion of methoxamine, electrical stimulation of the perivascular sensory outflow ( $0.56\text{--}5.6$  Hz) and i.v. bolus injections of exogenous  $\alpha$ -CGRP ( $0.1\text{--}1$   $\mu\text{g/kg}$ ) resulted in, respectively, frequency-dependent (Fig. 2a) and dose-dependent (Fig. 2b) vasodepressor responses. The former appeared about 10 s after starting each electrical stimulus and reached a maximum 1 min after the stimulus had ended (see Fig. 1); whereas the latter were immediate and reached a maximum 1 min after the corresponding i.v. injection had been given (data not shown). Consistent with previous findings (Taguchi et al., 1992b; Villalón et al., 2008), the vasodepressor responses produced by both methods: (i) returned to baseline levels after 5–10 min; and (ii) were not accompanied by significant changes ( $P > 0.05$ ) in heart rate (see Fig. 1), a finding that confirms, once again, the selectivity of stimulation at the vascular level.

Moreover, both electrically-induced (Fig. 2a) and  $\alpha$ -CGRP-induced (Fig. 2b) vasodepressor responses remained without significant changes ( $P > 0.05$ ) during the continuous i.v. infusions of saline ( $0.02$  ml/min; vehicle to dissolve sumatriptan, magnesium valproate and propranolol) or 10% propylene glycol in saline ( $0.02$  ml/min; vehicle to dissolve ergotamine and dihydroergotamine).

### Effect of sumatriptan, ergotamine, dihydroergotamine, valproate or propranolol on the electrically-induced vasodepressor responses

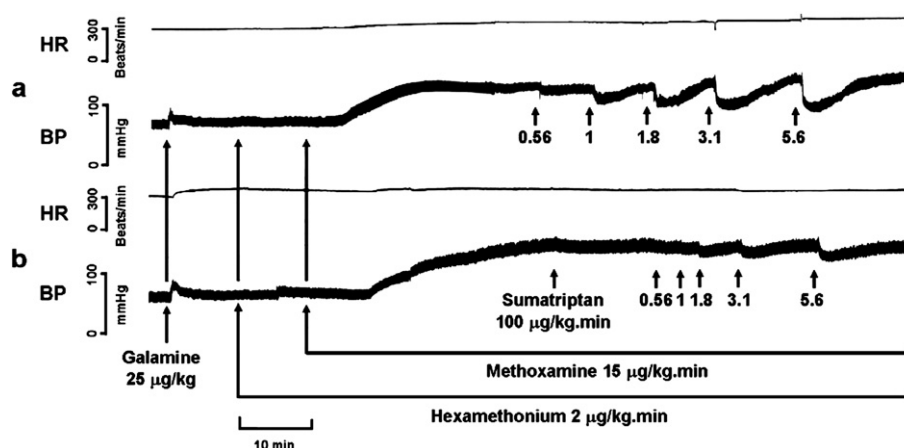
Fig. 3 shows that the infusions of sumatriptan ( $1, 10$  and  $100$   $\mu\text{g/kg.min}$ ; Fig. 3a and b), ergotamine ( $0.18, 0.31$  and  $0.56$   $\mu\text{g/kg.min}$ ; Fig. 3c and d) or dihydroergotamine ( $1, 3$  and  $10$   $\mu\text{g/kg.min}$ ; Fig. 3e and f), but

**Table 1**

Values of diastolic blood pressure and heart rate during the continuous i.v. infusion of methoxamine ( $15\text{--}20$   $\mu\text{g/kg.min}$ ) before and 10 min after the different treatments (nothing or continuous i.v. infusions of several compounds) had been started (during the infusion of methoxamine) in the animals receiving electrical stimulation

| Treatment         | Dose ( $\mu\text{g/kg.min}$ ) | n | Diastolic blood pressure (mmHg) |              | Heart Rate (beats/min) |              |
|-------------------|-------------------------------|---|---------------------------------|--------------|------------------------|--------------|
|                   |                               |   | Before                          | After        | Before                 | After        |
| Control           | –                             | 6 | $121 \pm 8$                     | $128 \pm 5$  | $286 \pm 15$           | $301 \pm 10$ |
| Saline            | $0.02$ ml/min                 | 6 | $112 \pm 6$                     | $118 \pm 7$  | $281 \pm 18$           | $290 \pm 19$ |
| Propylene glycol  | $0.02$ ml/min                 | 6 | $128 \pm 7$                     | $138 \pm 7$  | $288 \pm 16$           | $298 \pm 17$ |
| Sumatriptan       | 1                             | 6 | $131 \pm 17$                    | $137 \pm 15$ | $387 \pm 25$           | $399 \pm 30$ |
| Sumatriptan       | 10                            | 6 | $147 \pm 13$                    | $160 \pm 8$  | $344 \pm 14$           | $355 \pm 8$  |
| Sumatriptan       | 100                           | 6 | $157 \pm 2$                     | $159 \pm 6$  | $309 \pm 28$           | $310 \pm 34$ |
| Ergotamine        | 0.18                          | 6 | $139 \pm 10$                    | $147 \pm 10$ | $316 \pm 9$            | $317 \pm 13$ |
| Ergotamine        | 0.31                          | 6 | $132 \pm 9$                     | $129 \pm 11$ | $315 \pm 8$            | $316 \pm 9$  |
| Ergotamine        | 0.56                          | 6 | $140 \pm 14$                    | $136 \pm 12$ | $312 \pm 6$            | $315 \pm 2$  |
| Dihydroergotamine | 1                             | 6 | $128 \pm 9$                     | $137 \pm 9$  | $299 \pm 18$           | $302 \pm 10$ |
| Dihydroergotamine | 3                             | 6 | $116 \pm 6$                     | $121 \pm 8$  | $272 \pm 10$           | $273 \pm 13$ |
| Dihydroergotamine | 10                            | 6 | $134 \pm 6$                     | $129 \pm 8$  | $310 \pm 26$           | $306 \pm 19$ |
| Mg valproate      | 1000                          | 6 | $148 \pm 18$                    | $155 \pm 18$ | $290 \pm 3$            | $315 \pm 31$ |
| Mg valproate      | 1800                          | 6 | $129 \pm 10$                    | $129 \pm 13$ | $300 \pm 15$           | $305 \pm 19$ |
| Propranolol       | 100                           | 6 | $121 \pm 2$                     | $128 \pm 4$  | $326 \pm 8$            | $321 \pm 9$  |
| Propranolol       | 300                           | 6 | $116 \pm 6$                     | $113 \pm 9$  | $340 \pm 12$           | $308 \pm 25$ |

The above treatments produced no significant changes ( $P > 0.05$ ) in blood pressure or heart rate.



**Fig. 1.** Original experimental tracings illustrating vasodepressor responses produced by electrical stimulation of the sensory CGRPergic outflow during the continuous infusions of: (a) methoxamine (15–20 µg/kg.min) or (b) methoxamine plus sumatriptan (100 µg/kg.min) in pithed rats. Note that the vasodepressor responses obtained with the infusions of methoxamine plus sumatriptan are smaller in comparison with the single infusion of methoxamine. Similar results were obtained with the total of experiments ( $n=6$  for methoxamine and methoxamine plus sumatriptan).

not those of magnesium valproate (1000 and 1800 µg/kg.min; Fig. 3g) or propranolol (100 and 300 µg/kg.min; Fig. 3h), produced an inhibition of the electrically-induced vasodepressor responses. This inhibition produced by the above acute antimigraine drugs was: (i) unrelated to changes in baseline diastolic blood pressure or heart rate (see Table 1); (ii) more pronounced at higher frequencies of stimulation (1.8, 3.1 and 5.6 Hz); (iii) practically dose-dependent with the corresponding first two doses (see Fig. 3a, c and e, respectively); and (iv) supramaximal when the corresponding third dose was infused (see Fig. 3b, d and f, respectively), i.e. did not significantly differ ( $P>0.05$ ) from that produced by the corresponding second dose.

On the basis of the above results, and considering that magnesium valproate and propranolol were inactive, the second infusion dose of sumatriptan (10 µg/kg.min), ergotamine (0.31 µg/kg.min) and dihydroergotamine (3 µg/kg.min) were chosen to further investigate whether they are capable of inhibiting the vasodepressor responses induced by i.v. bolus injections of exogenous  $\alpha$ -CGRP.

#### Effect of sumatriptan, ergotamine and dihydroergotamine on the vasodepressor responses induced by exogenous $\alpha$ -CGRP

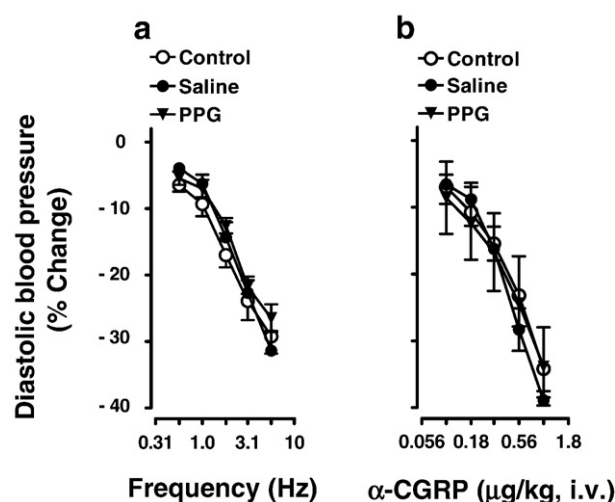
As shown in Fig. 4, during the continuous i.v. infusions of sumatriptan (10 µg/kg.min), ergotamine (0.31 µg/kg.min) and dihydroergotamine (3 µg/kg.min) the vasodepressor responses produced by i.v. bolus injections of exogenous  $\alpha$ -CGRP (0.1, 0.18, 0.31, 0.56 and 1 µg/kg) remained essentially unchanged when compared to the

corresponding control vasodepressor responses ( $P>0.05$ ). Likewise, there were no significant changes in the time-course of these responses in all subgroups. Note that the above infusion doses did inhibit the electrically-induced vasodepressor responses (see Fig. 3).

## Discussion

### General

Although sumatriptan (Williamson et al., 1997; Limmroth et al., 2001; Gupta et al., 2006) and dihydroergotamine (Buzzi et al., 1991) have been shown to inhibit neurogenic vasodilatation associated with trigeminal release of CGRP, the present study has analyzed whether a number of acute and prophylactic antimigraine drugs can produce a peripheral inhibition of the CGRPergic perivascular sensory outflow in pithed rats. It is important to note that, for this purpose, only one stimulus–response curve could be carried out per animal since tachyphylaxis was observed when eliciting a second stimulus–response curve (Villalón et al., 2008). This phenomenon may involve



**Fig. 2.** Vasodepressor responses produced by electrical stimulation (a) or i.v. bolus injections of exogenous  $\alpha$ -CGRP (b) during a continuous i.v. infusion of methoxamine (15–20 µg/kg.min) followed by: (i) nothing (control group); (ii) saline (0.02 ml/min); or (iii) 10% propylene glycol (PPG; 0.02 ml/min) ( $n=6$  each group).

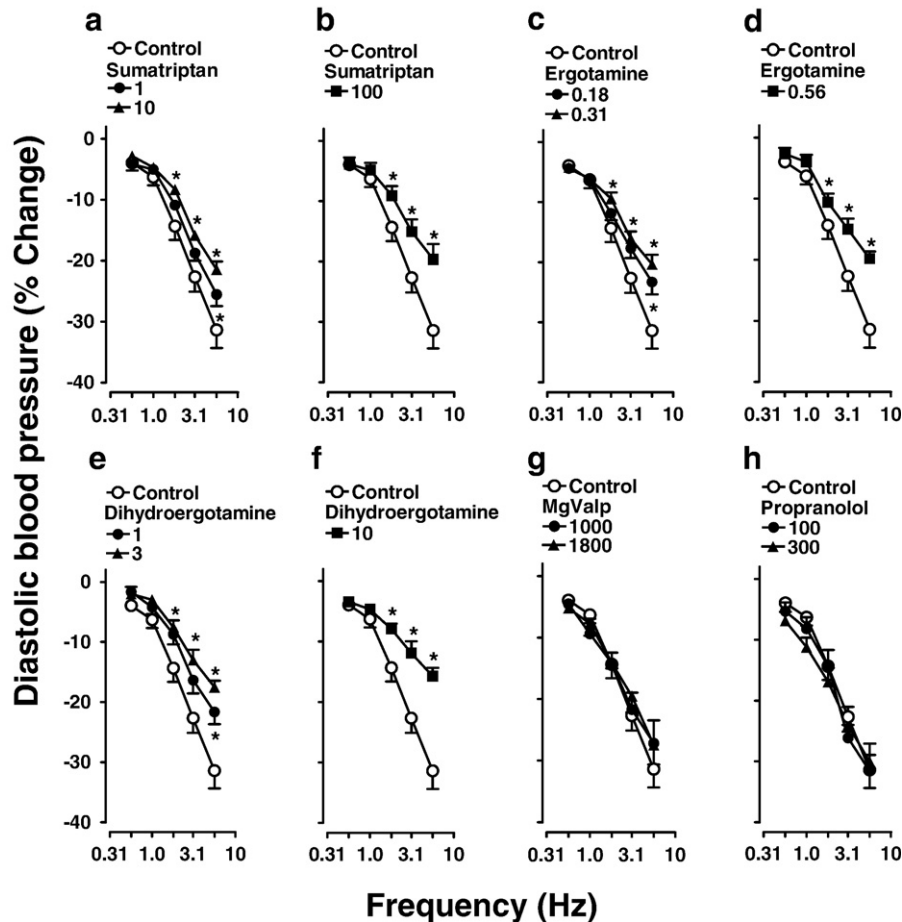
**Table 2**

Values of diastolic blood pressure and heart rate during the continuous i.v. infusion of methoxamine (15–20 µg/kg.min) before and 10 min after the different treatments (nothing or continuous i.v. infusions of several compounds) had been started (during the infusion of methoxamine) in the animals receiving i.v. bolus injections of  $\alpha$ -CGRP

| Treatment         | Dose (µg/kg.min) | n | Diastolic blood pressure (mmHg) |        | Heart rate (beats/min) |        |
|-------------------|------------------|---|---------------------------------|--------|------------------------|--------|
|                   |                  |   | Before                          | After  | Before                 | After  |
| Control           | –                | 6 | 132±11                          | 137±11 | 326±11                 | 325±9  |
| Saline            | 0.02 ml/min      | 6 | 137±14                          | 140±16 | 267±22                 | 269±17 |
| Propylene glycol  | 0.02 ml/min      | 6 | 164±4                           | 168±3  | 291±19                 | 291±23 |
| Sumatriptan       | 10               | 6 | 142±10                          | 153±13 | 324±30                 | 335±17 |
| Ergotamine        | 0.31             | 6 | 177±8                           | 187±7  | 344±27                 | 337±41 |
| Dihydroergotamine | 3                | 6 | 140±5                           | 145±5  | 286±42                 | 286±40 |

The above treatments produced no significant changes ( $P>0.05$ ) in blood pressure or heart rate.





**Fig. 3.** Effect of continuous i.v. infusions of sumatriptan (a for 1 and 10  $\mu\text{g/kg.min}$  and b for 100  $\mu\text{g/kg.min}$ ); ergotamine (c for 0.18 and 0.31  $\mu\text{g/kg.min}$  and d for 0.56  $\mu\text{g/kg.min}$ ); dihydroergotamine (e for 1 and 3  $\mu\text{g/kg.min}$  and f for 10  $\mu\text{g/kg.min}$ ); magnesium valproate (MgValp; g for 1000 and 1800  $\mu\text{g/kg.min}$ ); or propranolol (h for 100 and 300  $\mu\text{g/kg.min}$ ) ( $n=6$  for each dose) on the electrically-induced vasodepressor responses produced during a continuous i.v. infusion of methoxamine (15–20  $\mu\text{g/kg.min}$ ). The control group represents that receiving only a continuous i.v. infusion of methoxamine (15–20  $\mu\text{g/kg.min}$ ; shown for comparison). \*  $P<0.05$  vs. the corresponding control response.

depletion of neuronal CGRP, uncoupling from the G protein, sequestration and/or receptor down-regulation of CGRP receptors (see Buxton, 2006). However, our experimental model did not allow us to assess the extent to which each of these mechanisms is involved.

Apart from the implications discussed below, our study shows that the acute (i.e. sumatriptan, ergotamine and dihydroergotamine), but not the prophylactic (i.e. magnesium valproate and propranolol), antimigraine drugs can inhibit the vasodepressor responses induced by stimulation of the perivascular sensory outflow (Fig. 3). Since the second infusion dose of these acute antimigraine drugs failed to inhibit exogenous  $\alpha$ -CGRP-induced vasodepressor responses, as previously shown for clonidine (Villalón et al., 2008), a prejunctional inhibition of the perivascular sensory outflow is implied.

On the other hand, the two highest doses of sumatriptan (10 and 100  $\mu\text{g/kg.min}$ ; Fig. 3a and b), ergotamine (0.31 and 0.56  $\mu\text{g/kg.min}$ ; Fig. 3c and d) and dihydroergotamine (3 and 10  $\mu\text{g/kg.min}$ ; Fig. 3e and f) produced the same inhibition on the electrically-induced vasodepressor responses. Hence, the preceding infusion dose was chosen to analyze their effects on exogenous  $\alpha$ -CGRP-induced vasodepressor responses.

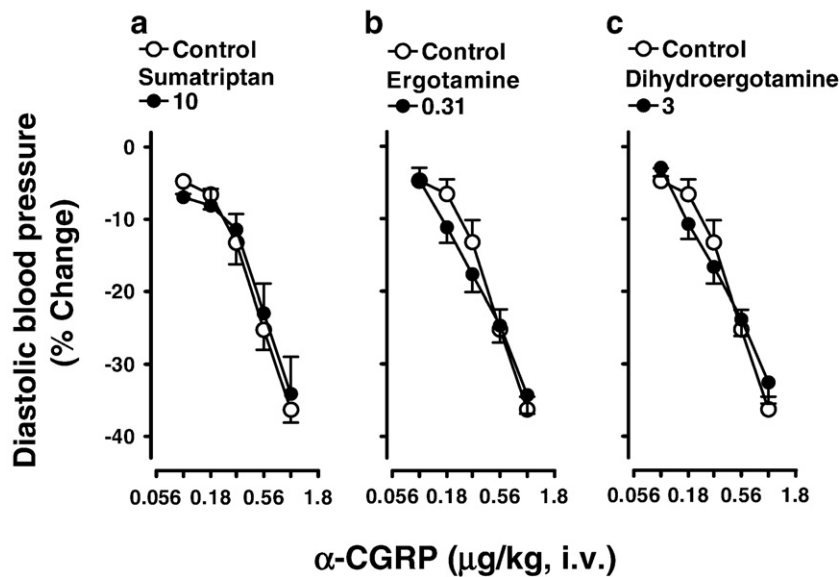
#### Systemic haemodynamic effects produced by the different treatments

The sustained increase in diastolic blood pressure by methoxamine (around 130 mmHg; see Table 1), resulting from an increase in peripheral vascular resistance (see Westfall and Westfall, 2006b), can be mainly attributed to stimulation of vascular  $\alpha_1$ -adrenoceptors (Decker et al., 1984). In contrast, we have no clear-cut explanation for

the increase in heart rate produced by the infusion of methoxamine (see “Haemodynamic variables” in Results section), which does not activate  $\beta$ -adrenoceptors (see Westfall and Westfall, 2006b). These effects are most likely to be drug-induced as a similar infusion schedule of saline in pithed rats did not significantly change heart rate (Sánchez-López et al., 2003).

It is noteworthy that the subsequent infusions of sumatriptan, ergotamine, dihydroergotamine, magnesium valproate or propranolol failed to further modify the increase in diastolic blood pressure during the infusion of methoxamine (see Table 1). In this respect, the acute antimigraine drugs sumatriptan, dihydroergotamine and ergotamine have been reported to produce cranial vasoconstriction by activation of, respectively: (i) 5-HT<sub>1B</sub> receptors (De Vries et al., 1998); (ii) 5-HT<sub>1B</sub> and  $\alpha_{2A/2C}$ -adrenoceptors (Villalón et al., 2004); and (iii) 5-HT<sub>1B</sub>,  $\alpha_{2A/2C}$ - and, to a lesser extent,  $\alpha_1$ -adrenoceptors (Valdivia et al., 2004). Moreover, stimulation of  $\alpha_1/\alpha_2$ -adrenoceptors, 5-HT<sub>2</sub> and, to a much lesser extent, 5-HT<sub>1B</sub> receptors in the systemic vasculature produces vasoconstriction and vasopressor responses (see Villalón and Centurión, 2007; Willems et al., 2003).

However, under our experimental conditions, the lack of effect of sumatriptan, dihydroergotamine and ergotamine on methoxamine-induced increase in diastolic blood pressure (Tables 1 and 2) may be due to the fact that this increase was supramaximal, as previously reported (Villalón et al., 2008). This condition could, in turn, have precluded further significant increases in this parameter when the above vasoconstrictor agents were infused, but allowed us to observe their prejunctional inhibitory actions on the perivascular sensory outflow (Fig. 3).



**Fig. 4.** Effect of continuous i.v. infusions of sumatriptan (a; 10  $\mu\text{g/kg.min}$ ), ergotamine (b; 0.31  $\mu\text{g/kg.min}$ ) or dihydroergotamine (c; 3  $\mu\text{g/kg.min}$ ) on the vasodepressor responses produced by i.v. bolus injections of exogenous  $\alpha$ -CGRP during a continuous i.v. infusion of methoxamine (15–20  $\mu\text{g/kg.min}$ ) ( $n=6$  for each group).

In addition, our study reconfirms that electrical stimulation of the perivascular sensory outflow (in the presence of hexamethonium and methoxamine) and i.v. bolus injections of  $\alpha$ -CGRP produce, respectively, frequency-dependent (Fig. 2a) and dose-dependent (Fig. 2b) vasodepressor responses. Both vasodepressor responses: (i) were highly reproducible as they remained practically unaltered during the infusions of saline or 10% propylene glycol; and (ii) may be attributed to the potent vasodilator actions of CGRP (Brain et al., 1985) and to the high density of CGRP receptors in the intima and media layers of systemic blood vessels (see Wimalawansa, 1996).

#### *Inhibition of the electrically-induced, but not of the exogenous $\alpha$ -CGRP-induced, vasodepressor responses by sumatriptan, ergotamine or dihydroergotamine*

The inhibition of the electrically-induced vasodepressor responses (and, consequently, of the perivascular sensory outflow) by sumatriptan, ergotamine and dihydroergotamine (Fig. 3) is consistent with other findings reported in rodents. Thus, sumatriptan has been shown to produce a 5-HT<sub>1B/1D</sub> receptor-mediated prejunctional inhibitory action on: (i) neurogenic inflammation induced by nerve stimulation or capsaicin (Carmichael et al., 2008); (ii) dural vasodilatation induced by perivascular trigeminal stimulation (Gupta et al., 2006); and (iii) cardiac sympathetic outflow (Sánchez-López et al., 2003, 2004). Furthermore, ergotamine and dihydroergotamine have been shown to produce a prejunctional inhibitory action on: (i) trigeminal outflow (Saito et al., 1988); (ii) dural plasma protein extravasation associated with release of neuropeptides (Markowitz et al., 1988); and (iii) cardiac sympathetic outflow (Roquebert and Grenié, 1986). Indeed, functional  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors can be activated by ergotamine (Valdivia et al., 2004) or dihydroergotamine (Villalón et al., 2004), and activation of these receptors has also been associated with inhibition of neurotransmitter release, including neuropeptides (Boehm and Kubista, 2002; Sánchez-López et al., 2003, 2004; Villalón et al., 2008). The above findings, coupled to the lack of effect of sumatriptan, ergotamine or dihydroergotamine on the vasodepressor responses produced by i.v. bolus injections of exogenous  $\alpha$ -CGRP (see Fig. 4), reinforce our view that these acute antimigraine drugs attenuate the electrically-induced vasodepressor responses mainly by a prejunctional inhibitory action on the perivascular sensory outflow,

as previously shown for clonidine (Villalón et al., 2008). Clearly, additional experiments, which fall beyond the scope of the present investigation, will be required to further investigate the pharmacological profile of the mechanisms involved in the prejunctional inhibitory action of the above acute antimigraine drugs.

#### *Failure of valproate and propranolol to inhibit the perivascular sensory outflow*

There currently exists a wide array of prophylactic antimigraine agents with proven efficacy (Arulmozhi et al., 2006; Villalón et al., 2003), including the anticonvulsant drug magnesium valproate (Sørensen, 1988) and the  $\beta$ -adrenoceptor antagonist propranolol (Diener, 2003). Unfortunately, their mechanisms of antimigraine action are still elusive and no study has evaluated whether these drugs can inhibit the perivascular sensory outflow. With this in mind, our study shows that, in contrast to the acute antimigraine drugs (see above), magnesium valproate (Fig. 3g) and propranolol (Fig. 3h) failed to significantly modify the electrically-induced vasodepressor responses even at very high (1800 and 300  $\mu\text{g/kg.min}$ , respectively) doses. Our results with propranolol confirm and extend previous findings showing the lack of effect of this  $\beta$ -blocker on the vasodepressor sensory CGRPergic outflow (Taguchi et al., 1992b). However, these studies were carried out using a single frequency of stimulation (4 Hz) and only one dose of propranolol (500  $\mu\text{g/kg}$ , i.v.). For this reason, we considered it important to reanalyze the effect of propranolol on the vasodepressor sensory outflow, using several frequencies of stimulation (0.56–5.6 Hz) and several doses of propranolol (100 and 300  $\mu\text{g/kg.min}$  i.v.), which failed to inhibit the CGRPergic sensory outflow (see Fig. 3h), whereas 500 and 1000  $\mu\text{g/kg.min}$  produced an immediate cardiac arrest followed by death (not shown).

Moreover, our results with valproate are at variance with previous studies showing that valproate inhibited the neurogenic inflammation related to substance P release (Cutrer et al., 1995; Lee et al., 1995). This apparent discrepancy may be at least related to differences in: (i) the experimental conditions; and (ii) the nature of the neuropeptide being released. On the basis of the above findings, we decided not to further analyze the effects of magnesium valproate and propranolol on the vasodepressor responses produced by i.v. bolus injections of exogenous  $\alpha$ -CGRP.

## Conclusion

The above results, taken together, suggest that sumatriptan (10 µg/kg.min), ergotamine (0.31 µg/kg.min) and dihydroergotamine (3 µg/kg.min), but not magnesium valproate or propranolol, attenuate the electrically-induced vasodepressor responses mainly by a prejunctional inhibitory action on the perivascular sensory outflow. This inhibitory effect may be of particular relevance when considering potential cardiovascular adverse effects by acute antimigraine drugs.

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